

ITA0520 – Computer Vision For Image Analysis

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Experiment 16:

Implement a Sobel algorithm using Open CV to filter the input image.

Code:

```
import cv2

import numpy as np

image = cv2.imread("Person.jpg")

if image is None:

    print(" ❌ Error: Image not found!")

else:

    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    blur = cv2.GaussianBlur(gray, (5, 5), 0)

    sobel_x = cv2.Sobel(blur, cv2.CV_64F, 1, 0, ksize=3)

    sobel_y = cv2.Sobel(blur, cv2.CV_64F, 0, 1, ksize=3)

    sobel_x = cv2.convertScaleAbs(sobel_x)

    sobel_y = cv2.convertScaleAbs(sobel_y)

    sobel_combined = cv2.addWeighted(sobel_x, 0.5, sobel_y, 0.5, 0)

    cv2.imshow("Original Image", image)

    cv2.imshow("Sobel X", sobel_x)

    cv2.imshow("Sobel Y", sobel_y)

    cv2.imshow("Sobel Combined", sobel_combined)

    cv2.waitKey(0)

    cv2.destroyAllWindows()
```

Input:



Output:

